

Desmos Mastery - Day 7

14 original questions | Calculator permitted

Complete the questions before turning to the answer explanations. Mark any guess, even if it turns out to be correct. For math, show your setup and check the requested quantity. For reading, identify the exact supporting detail; for grammar, identify the clause or subject.

These questions were authored for this workbook from the lesson topics. They are not official SAT questions or copies of the course's paid practice.

Isolation | 7 questions

[Lesson video: Isolation](#)

1. Which equation expresses I in terms of Q , A , and L , where $L \neq 0$?

$$Q = \frac{A - I}{L}$$

- A) $I = A - QL$
- B) $I = QL - A$
- C) $I = (A - Q)/L$
- D) $I = A/Q - L$

2. A sphere has volume V and positive radius r . Which expression gives r ?

$$V = \frac{4}{3}\pi r^3$$

- A) $\sqrt{(3V/(4\pi))}$
- B) $\sqrt[3]{(3V/(4\pi))}$
- C) $4\pi V/3$
- D) $3V/(4\pi)$

3. Which expression gives t in the equation below when $v + u \neq 0$?

$$s = \frac{u + v}{2}t$$

- A) $2s/(v - u)$
- B) $2s/(v + u)$
- C) $s(v + u)/2$
- D) $s/(2v + 2u)$

4. Which equation expresses x in terms of y , where $y \neq 3$?

$$y = \frac{3x+5}{x-2}$$

- A) $x = (2y + 5)/(y - 3)$
- B) $x = (2y - 5)/(y + 3)$
- C) $x = (3y - 5)/(y - 2)$
- D) $x = (y - 3)/(2y + 5)$

5. Which equation expresses h in terms of A , a , and b when $a + b \neq 0$?

$$A = \frac{a+b}{2}h$$

- A) $h = 2A/(a + b)$
- B) $h = A/(2a + 2b)$
- C) $h = 2A(a + b)$
- D) $h = (a + b)/(2A)$

6. Positive numbers x and y satisfy $y = 5x^2$. When $y = 180$, what is x ?

Answer: _____

7. Which formula expresses b in terms of c and a , where $c \neq 1$?

$$C = \frac{a+b}{b}$$

- A) $b = a/(c - 1)$
- B) $b = a/(c + 1)$
- C) $b = a(c - 1)$
- D) $b = (c - 1)/a$

Factoring | 7 questions

[Lesson video: Factoring](#)

8. The polynomial $x^2 + kx - 24$ has $x - 3$ as a factor. What is k ?

Answer: _____

9. The polynomial below has $x + 2$ as a factor. What is a ?

$$2x^2 + ax - 18$$

Answer: _____

10. Which polynomial has $x - 4$ as a factor?

- A) $x^2 - 4x + 4$
- B) $x^2 - 7x + 12$
- C) $x^2 + 7x + 12$
- D) $x^2 - 2x - 12$

11. The polynomial $(x - 5)(x + 1)$ is written as $x^2 + bx + c$. What is $b - c$?

Answer: _____

12. For a positive constant r , $x - r$ is a factor of $x^2 - 49$. What is r ?

Answer: _____

13. The polynomials below share the factor $x - 2$. What is $m + n$?

$$x^2 + mx + 6, \quad 3x^2 - 5x + n$$

Answer: _____

14. The polynomial $x^2 - 10x + c$ has a repeated real root. What is c ?

Answer: _____

Answers & explanations

Count correct answers, but also review correct guesses. Redo missed questions after a day without looking at the solution.

1: A 2: B 3: B 4: A 5: A 6: 6 7: A 8: 5 9: -5 10: B 11: 1 12: 7 13: -7 14: 25

Why each answer works

- 1. Answer: A.** Multiply by L to obtain $QL = A - I$. Then $I = A - QL$.
- 2. Answer: B.** From $V = (4/3)\pi r^3$, obtain $r^3 = 3V/(4\pi)$ and take the positive cube root.
- 3. Answer: B.** Multiply by 2 and divide by $u + v$: $t = 2s/(u + v)$. The denominator must be nonzero.
- 4. Answer: A.** Multiply $y(x - 2) = 3x + 5$. Collect x -terms: $x(y - 3) = 2y + 5$, then divide by $y - 3$.
- 5. Answer: A.** The trapezoid area formula is $A = (a + b)h/2$. Multiplying by 2 and dividing by $a + b$ isolates h .
- 6. Answer: 6.** $x^2 = 36$, and positivity selects $x = 6$. Without positivity, both 6 and -6 would solve the equation.
- 7. Answer: A.** Multiply by b : $cb = a + b$. Then $b(c - 1) = a$, so $b = a/(c - 1)$, with the original requirement $b \neq 0$ still applying.
- 8. Answer: 5.** The factor theorem gives $3^2 + 3k - 24 = 0$, so $3k = 15$ and $k = 5$.
- 9. Answer: -5.** Substitute $x = -2$: $2(4) - 2a - 18 = 0$. Therefore $-2a = 10$ and $a = -5$.
- 10. Answer: B.** At $x = 4$, the second polynomial is $16 - 28 + 12 = 0$; it factors as $(x - 3)(x - 4)$. The other values are 4, 56, and -4 .
- 11. Answer: 1.** Expand to $x^2 - 4x - 5$. Thus $b = -4$ and $c = -5$, so $b - c = 1$.
- 12. Answer: 7.** The roots are -7 and 7 . Since $r > 0$, the factor $x - r$ must be $x - 7$.
- 13. Answer: -7.** For the first, $4 + 2m + 6 = 0$ gives $m = -5$. For the second, $12 - 10 + n = 0$ gives $n = -2$. Their sum is -7 .
- 14. Answer: 25.** A repeated root requires completing the square with no remainder: $x^2 - 10x + 25 = (x - 5)^2$. Equivalently, the discriminant $100 - 4c$ is zero.

For each missed or guessed question, record the skill, the condition you overlooked, and one specific action for next time.